

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H1



**ST. ANDREW'S JUNIOR COLLEGE
2021 JC2 PRELIMINARY EXAMINATIONS**

H1 BIOLOGY

8876/02

STRUCTURED QNS AND ESSAY

Monday

30th August 2021

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagram, graph.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

Write your answers in the spaces provided on the question paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiners' Use	
STQ	
1	/10
2	/20
3	/15
4 or 5	/15
Total	/60

Conceptual error (C)	Data Quoting (D)	Expression (E)	Misreading the question (Q)

This document consists of **20** printed pages.

[Turn over

Section A

Answer **all** the questions in this section.

1(a) T-helper lymphocytes and Leydig cells are two types of mammalian cells. The main role of T-helper lymphocytes and Leydig cells is to synthesise and secrete cell-signalling molecules (which are molecules that are released by one cell. Such cell-signalling molecules, e.g cytokines and testosterone, recognise and bind to receptor on the cell surface membrane of another cell where they cause a cellular response).

- T-helper lymphocytes synthesise proteins known as cytokines.
- Leydig cells synthesise the steroid (lipid) hormone testosterone from cholesterol. Leydig cells also synthesise cholesterol.

Figure 1.1 shows part of a mammalian cell.

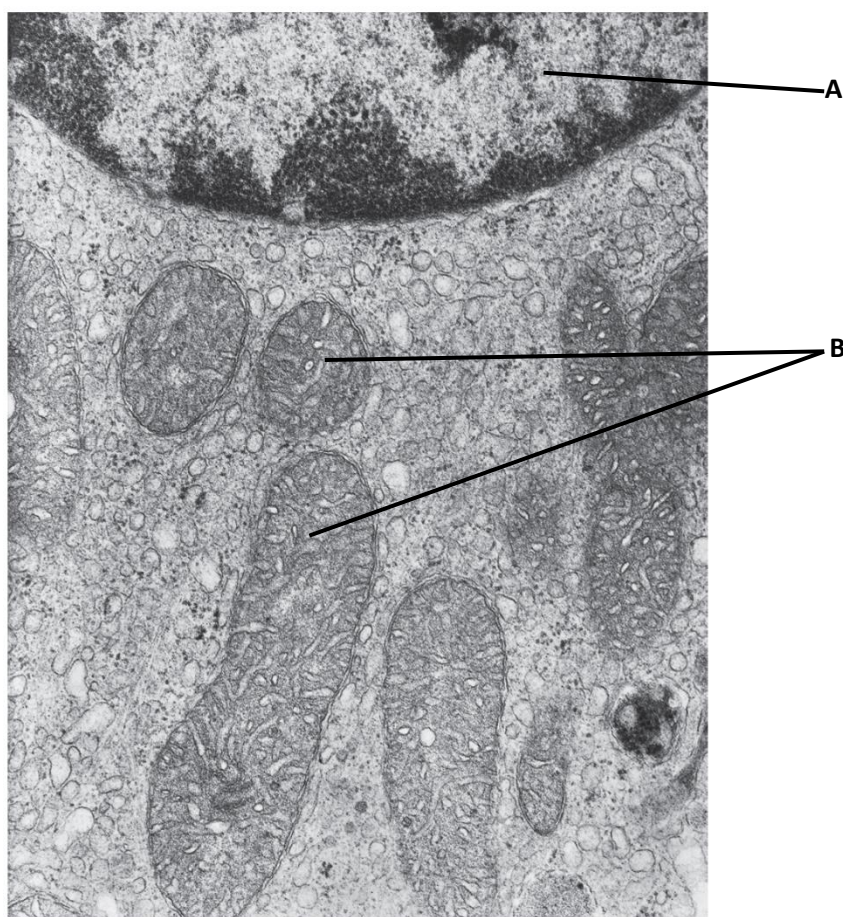


Fig. 1.1

(i) Identify organelles **A** and **B** on Fig. 1.1.

A:

B:

[1]

- (ii) Organelle **A** is involved in mitotic cell cycle. State the process that occurs during interphase to produce genetically identical daughter cells.

.....[1]

- (iii) Explain whether Fig. 1.1 shows part of a Leydig cell or part of a T-helper lymphocyte.

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[2]

- (b) Cell-surface receptors are membrane-anchored (integral) proteins that bind to external signalling molecules.

Such receptors are synthesised within the cell and brought to the cell surface membrane as shown in the Figure 1.2.

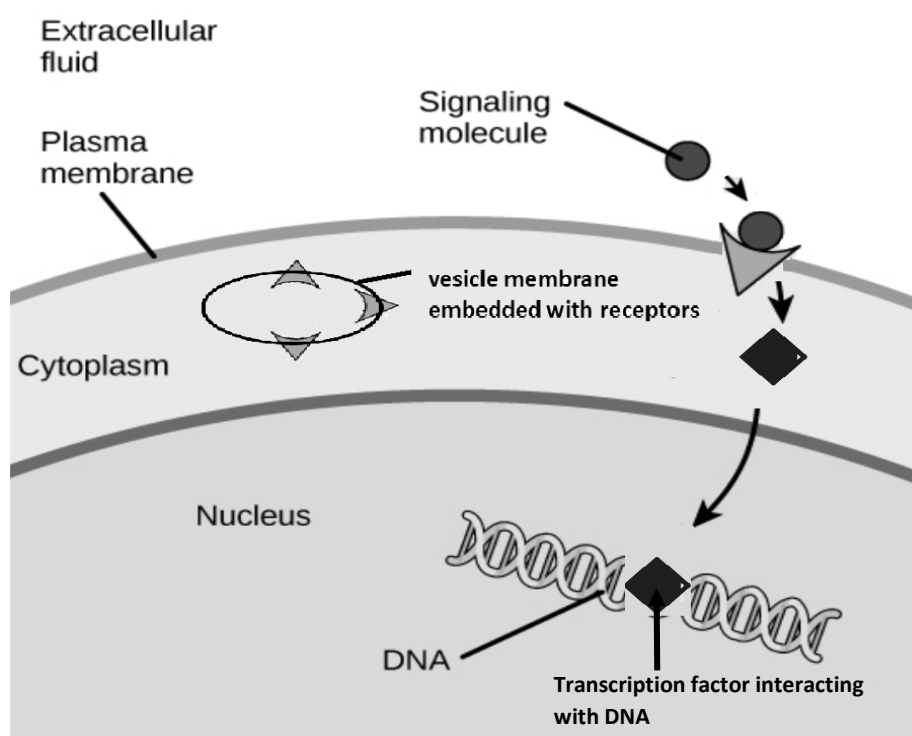


Fig. 1.2

With reference to Fig. 1.2, describe how the receptor is brought to and become part of the cell surface membrane.

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[2]

- (c) *Eichhornia crassipes*, a water hyacinth plant, is an invasive species found in lakes and rivers throughout Asia. They grow rapidly, producing many flowers and seeds. Each flower of this plant has six anthers (each anther is at one end of the filament) and one stigma as shown in Figure 1.3.

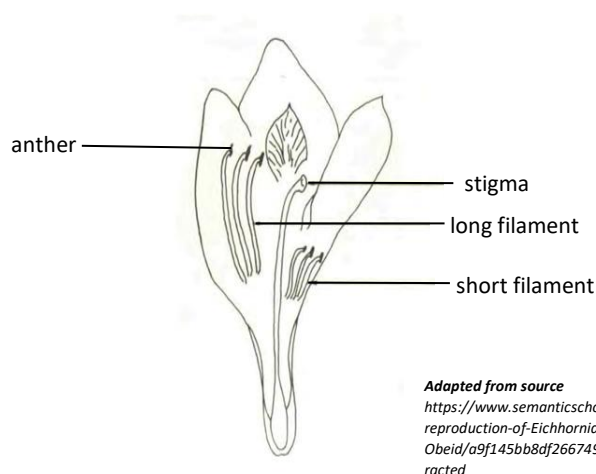


Fig. 1.3 showing spatial arrangement of stamen (comprising of both anther and filament) and stigma in *E. crassipes*.

Each anther was observed to release many pollen grains. After pollination, the pollen grains germinate after it comes into contact with the stigma, in response to the sugary fluid secreted by the mature stigma. A pollen tube grows out from each pollen grain. The pollen tube serves to transport the male gamete cells to the female gamete cells (ovules) in the flower.

The tip of the pollen tube is protected by an outer layer of pectin.

Pectin is a polysaccharide.

Starch grains, mitochondria and vesicles are present in the tip of the pollen tube.

Using the information given and your content knowledge on organelles and carbohydrates, describe how the production of pectin occurred at the tip of a pollen tube.

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.....[4]

[Total: 10]

2(a) Severe combined immunodeficiency (SCID) is a disease that affect the development and maturation of white blood cells known as lymphocytes.

There are two general types of SCID, both of which are caused by recessive alleles.

Fig. 2.1 shows a family pedigree including some people with a type of SCID.

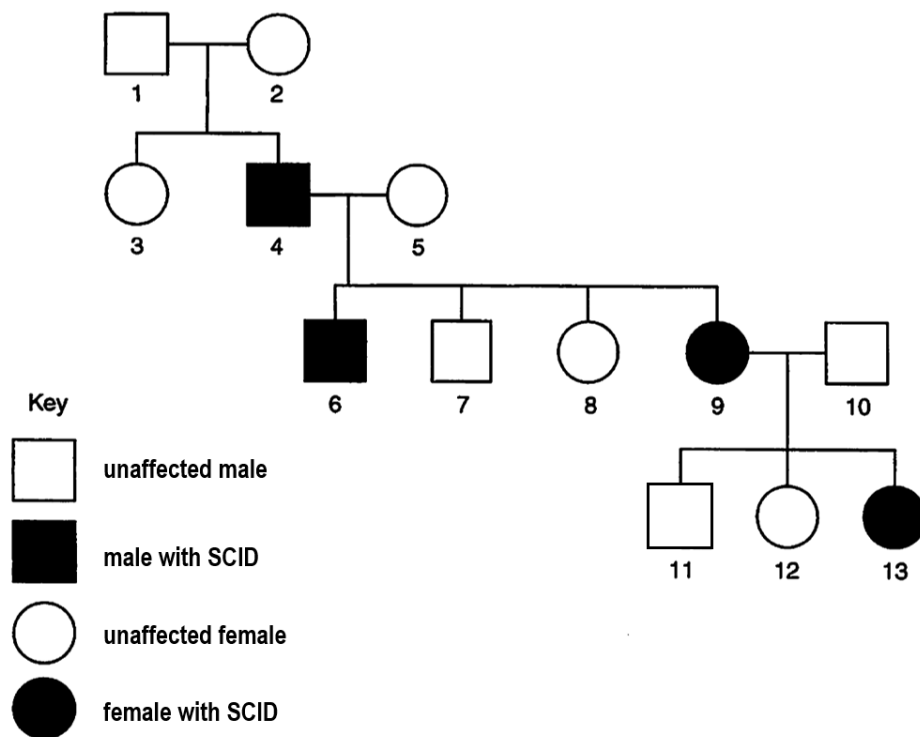


Fig. 2.1

(i) Explain what is meant by the term allele.

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.....[2]

(ii) Explain, with evidence in Fig. 2.1, whether the disease is X-linked.

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.....[3]

(b) In mice, the intensity of pigmentation of the fur is controlled by multiple alleles of a single gene.

The alleles are listed below in order of dominance, with **C** as the most dominant.

- **C** = full colour
- **C^{ch}** = chinchilla
- **C^h** = himalayan
- **C^p** = platinum
- **C^a** = albino

Eye colour in mice is controlled by two alleles of a single gene, **B/b**:

- allele **B** codes for black eyes
- allele **b** codes for red eyes.

A mouse with full colour fur and black eyes was crossed with a mouse with himalayan fur and black eyes. One of the offspring was albino with red eyes.

Using the symbols above, draw a genetic diagram to show the genotypes and phenotypes of the offspring of this cross.

(c) Figure 2.2 shows the nematode worm, *Caenorhabditis elegans*.



magnification $\times 250$

Fig. 2.2

Stem cells of *C. elegans* have been studied.

Figure 2.3 shows the change in mass of DNA per nucleus in a stem cell during one cell cycle. $1\text{pg} = 1 \times 10^{-12}\text{g}$.

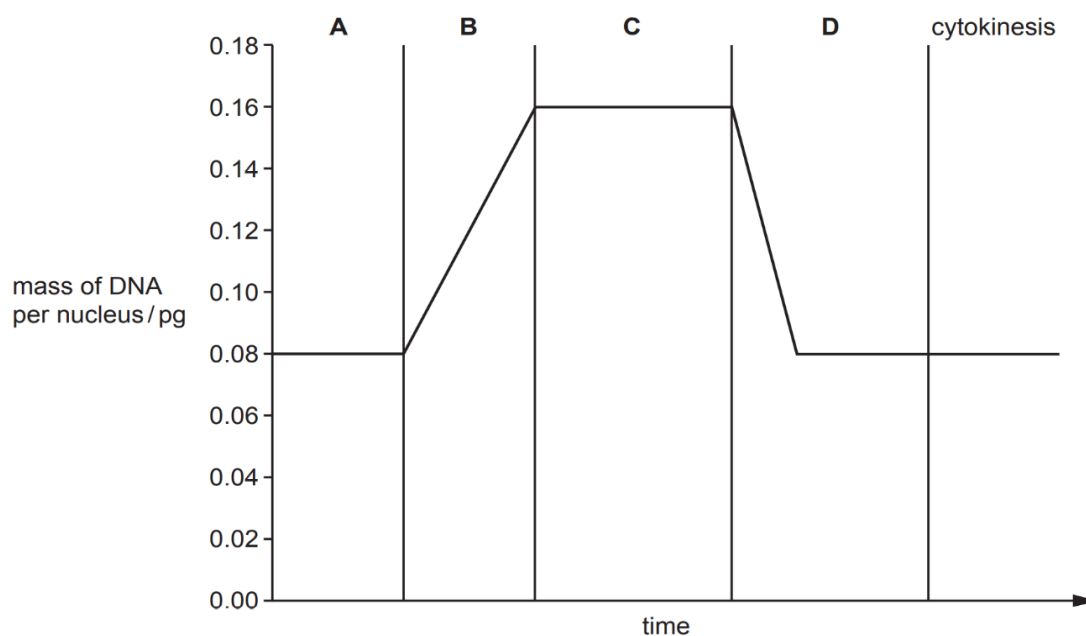


Fig. 2.3

(i) State the phases of interphase shown by **A** and **B** and the stage of mitosis shown by **D** in Fig. 2.3.

- A**
- B**
- D**

[1]

(ii) Outline what happens in a cell in stage D in preparation for cytokinesis.

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.....[2]

(iii) Immediately after cytokinesis, daughter cells are **not** identical even though they are genetically identical.

Suggest a reason why daughter cells are not identical immediately after cytokinesis.

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.....[1]

(iv) A stem cell of a female *C. elegans* has 12 chromosomes.

Complete Table 2.1 to show the number of nuclei within the stem cell in stages **A**, **B** and **D** of the cell cycle shown in Figure 2.3 and the number of chromosomes in each nucleus during these stages.

Table 2.1

	stage of the cell cycle		
	A	B	D
number of nuclei within the stem cell			
number of chromosomes in each nucleus			

[2]

- (d) Embryonic stem cells are extracted from human embryos 4-5 days after fertilisation, when the embryo consists of 50 to 150 cells.

Suggest one ethical concern about the use of embryonic stem cells.

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[1]

- (e) The tomato plant, *Solanum lycopersicum*, does not tolerate periods of drought (water shortage). Researchers have produced a tomato plant that has an improved tolerance of drought.

The nuclei of plants produce small lengths of RNA known as microRNAs.

MicroRNAs in guard cells have been shown to prevent the synthesis of some proteins.

The guard cells of the drought-tolerant tomato plants produced more microRNA molecules than the guard cells of the non-tolerant plants.

MicroRNA molecules do **not** prevent transcription but interact with messenger RNA (mRNA).

Suggest how this microRNA can interact with mRNA to prevent the production of proteins in guard cells of *S. lycopersicum*.

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[3]

[Total: 20]

3(a) ATP is often known as the universal 'energy currency' of the cell.

Most ATP are made in cells by membrane systems that create proton gradients by pumping protons from one compartment to another.

Figure 3.1 shows two of such membrane systems.

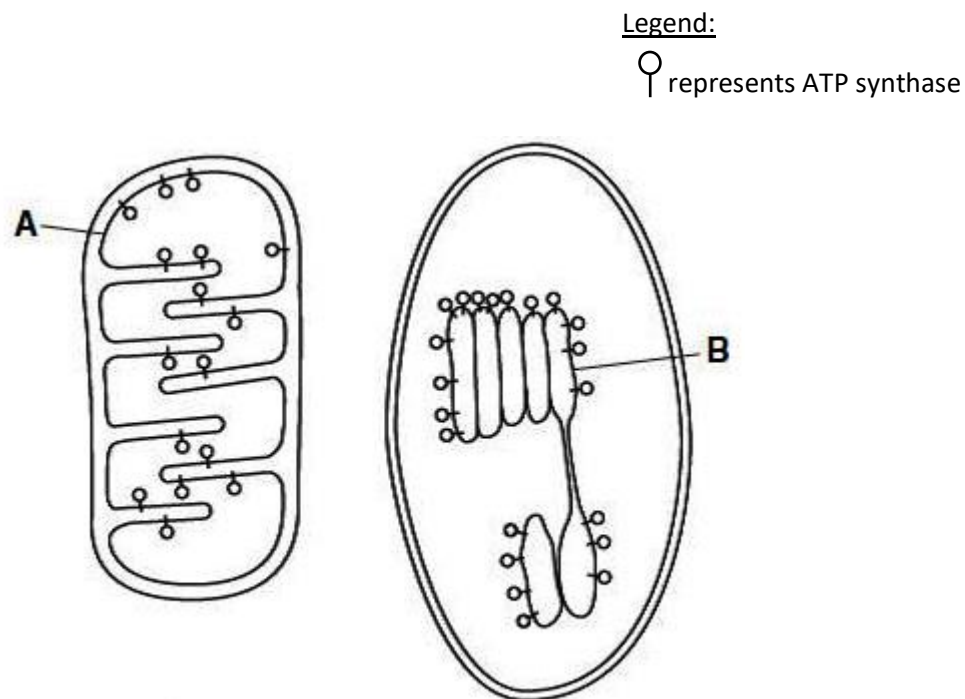


Fig. 3.1

- (i) Draw arrows onto **each** of the membrane systems in Fig. 3.1 to show the direction in which protons are pumped.

.....[1]

- (ii)** Outline how energy is generated for the pumping of protons across membrane system B.

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.....[3]

- (iii)** Explain why transmembrane proteins are necessary for protons to be pumped across the membrane systems.

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.....[2]

- (iv)** It is also known that there are cyclic pathways involving the use of enzymes occurring in both membrane systems.

State the names of the two cyclic pathways and outline the differences between them.

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.....[4]

- (b) Sehuencas water frogs are endemic to Bolivia. It was once believed that this frog was extinct in the wild. This was due to human activity and a disease caused by a fungus.

In 2008, there was only one Sehuencas water frog in captivity.

In 2018, five Sehuencas water frogs were discovered in a Bolivian mountain forest.

Three of these frogs were male. Scientists suggested that these five frogs were resistant to the fungal disease.

Explain how resistance to this fungus came about in a population of Sehuencas water frogs.

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.....[5]

[Total: 15]

Section B**Answer EITHER 4 or 5.**

Write your answers on lines provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)**, etc., as indicated in the question.

4 (a) Outline the process of DNA replication. [5]

(b) Explain how different types of gene mutation can affect the phenotype and outline the effects of the mutant alleles that cause sickle cell anemia on the phenotype of a person. [10]

[Total: 15]

5 (a) Many different polymers are formed in cells. A polymer is a chain of monomers. [8]

Describe the polymerisation of different types of biomolecules in a plant.

(b) Discuss how the complementary shapes between (bio)molecules enable eukaryotic cells to carry out their diverse processes. [7]

[Total: 15]

[illegible]

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END OF PAPER